FinSight Project 3 Report - Team 6

FinSight — Project 3 Deliverable Report

CS6.401 Software Engineering | Submitted by: Team 6 Project Repository: https://github.com/GojoSaturo0409/FinSight

Team Contributions

Member	Responsibility
Ananth	Frontend Development & UI Engineering
Lokesh	Analytics Pipeline & Data Visualization
Eshwar	Core Business Logic & Microservice Architecting
Rahul	Core Logic, API Integrations & Systems Integration
Ayush	Core Logic, API Integrations & Systems Integration

Task 1: Requirements and Subsystems

1.1 Functional Requirements

ID	Requirement	Architecturally Significant?	Rationale
FR-1	Users can register, log in, and receive JWT-based session tokens	Yes	Drives the Auth Service boundary; every other service depends on token validation

ID	Requirement	Architecturally Significant?	Rationale
FR-2	Users can ingest transactions from Plaid (bank), CSV uploads, or manual entry	Yes	Necessitates the Adapter & Factory patterns for multi-source normalization
FR-3	The system evaluates spending against user-defined budget limits per category	Yes	Core trigger for the Observer + Pub/Sub notification pipeline
FR-4	Budget breaches dispatch email (Mailjet) and push (Firebase) notifications	Yes	Requires asynchronous processing via Celery workers to avoid blocking the API
FR-5	Users can buy/sell investment equities and view portfolio value	No	Standard CRUD mapped to the Analytics Service
FR-6	AI-powered spending recommendations are generated from transaction history	No	Leverages Chain of Responsibility for sequential analysis handlers
FR-7	Monthly PDF reports are exportable with embedded charts	Yes	Requires WeasyPrint system-level dependencies inside Docker
FR-8	All monetary values are globally convertible across currencies (USD, EUR, INR…)	Yes	Drives the Chain of Responsibility fallback for live/cached/hardcoded rates

1.2 Non-Functional Requirements

ID	Requirement	Quality Attribute
NFR-1	Budget alerts must be delivered within 5 seconds of a breach event	**Performance**
NFR-2	Adding a new data source (e.g., Stripe) must require only a single new Adapter class	**Modifiability**

ID	Requirement	Quality Attribute
NFR-3	The notification pipeline must not block the main API response	**Availability**
NFR-4	Currency conversion must succeed even if the external API is offline	**Fault Tolerance**
NFR-5	Each user's data must be strictly isolated from other users	**Security**

1.3 Subsystem Overview

Subsystem	Technology	Role
Auth Service	FastAPI, JWT, bcrypt	User registration, login, password hashing, and token issuance
Transaction Service	FastAPI, Plaid SDK, SQLAlchemy	Multi-source transaction ingestion, normalization via Adapters, and currency conversion
Budget Service	FastAPI, Pika (RabbitMQ)	Budget limit management, threshold evaluation, and event publishing
Analytics Service	FastAPI, WeasyPrint, Chart.js	AI recommendations, investment portfolio CRUD, and PDF report generation

Subsystem	Technology	Role
Budget Worker	Celery, RabbitMQ	Asynchronous consumer that dispatches Mailjet emails and Firebase push notifications
API Gateway	Nginx	Unified reverse-proxy routing all `/api/*` traffic to appropriate microservices
Frontend SPA	React 18 (Vite), TypeScript	Interactive dashboard rendering charts, tables, budget sliders, and investment panels
Database	PostgreSQL 15	Centralized relational store for Users, Transactions, Budgets, and Investments
Message Broker	RabbitMQ	AMQP fanout exchange for decoupled event broadcasting between services

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Task 2: Architecture Framework

2.1 Stakeholder Identification (IEEE 42010)

Stakeholder	Concerns	Viewpoint	View
End User	Usability, data accuracy, real-time alerts	Functional	Dashboard UI, notification emails
Developer	Modifiability, testability, clear service boundaries	Development	C2 Container Diagram, codebase structure
DevOps	Deployability, containerization, monitoring	Deployment	Docker Compose topology, Nginx gateway
Security Auditor	Data isolation, credential management, JWT security	Security	Auth flow, `.env` separation, `.gitignore` rules
Course Evaluator	Correct application of SE patterns, working prototype	Logical	ADRs, UML diagrams, sequence diagrams

2.2 Architecture Decision Records (ADRs)

ADR-1: Adoption of Event-Driven Architecture with Celery & RabbitMQ

Field	Detail
Status	Accepted

Field	Detail
Context	Budget evaluations were originally synchronous inside the API loop. If Mailjet or Firebase responded slowly, the entire user request would block, degrading UX.
Decision	Introduce RabbitMQ as a fanout exchange and Celery as the background worker. The Budget API publishes lightweight `budget_event` messages; the Celery worker asynchronously consumes them and dispatches alerts.
Consequences	API response times became independent of third-party latency. Trade-off: adds operational complexity (RabbitMQ container, Celery process).

ADR-2: Adapter Pattern for Multi-Source Data Ingestion

Field	Detail
Status	Accepted
Context	The system ingests data from Plaid (nested JSON with accounts & transactions), CSV files (flat rows), and manual UI forms (simple key-value). Handling all formats in a single router method was unmaintainable.
Decision	Define an `ITransactionSource` interface. Implement `PlaidAdapter`, `CSVAdapter`, and `ManualEntryAdapter` that each normalize raw data into a unified `Dict[str, Any]` schema before DB persistence.
Consequences	Adding a new source (e.g., Stripe) requires only one new class implementing `fetch_transactions()`. Zero changes to the router or DB layer.

ADR-3: Chain of Responsibility for Currency Conversion Fallbacks

Field	Detail
Status	Accepted
Context	External currency APIs (exchangerate-api.com) are subject to rate-limiting and downtime. A single point of failure in conversion would break transaction ingestion entirely.
Decision	Engineer a three-stage chain: (1) Check in-memory/Redis cache → (2) Query live API → (3) Fall back to hardcoded constant rates. Each handler either resolves the request or passes it to the next.
Consequences	100% fault tolerance for currency conversion. Trade-off: hardcoded rates may be slightly stale during extended API outages.

ADR-4: Observer Pattern for Notification Pipelines

Field	Detail
Status	Accepted
Context	Delivering budget alerts required importing Mailjet SDK, Firebase Admin SDK, and file-logging logic directly inside the `BudgetMonitor`, creating tight coupling.
Decision	Implement the Observer pattern. `BudgetMonitor` maintains a list of `Observer` instances (`EmailNotifier`, `InAppNotifier`, `LoggingObserver`). On threshold breach, it iterates and calls `observer.update(category, threshold, spend, level)`.
Consequences	Adding a new channel (e.g., Twilio SMS) requires zero modifications to `BudgetMonitor`; simply append a new observer.

ADR-5: Shared Database with Logical Service Separation

Field	Detail
Status	Accepted (Pragmatic Compromise)
Context	Pure microservice orthodoxy dictates separate databases per service. However, the Budget Service needs to join `transactions` with `budgets`, and Analytics needs access to all tables for report generation.
Decision	Use a single PostgreSQL instance with logical separation via SQLAlchemy ORM models scoped by `user_id`. Each service accesses only its relevant tables through filtered queries.
Consequences	Eliminates costly inter-service REST calls for data aggregation. Trade-off: services share a schema, which could lead to unintended coupling if not disciplined.

Task 3: Architectural Tactics and Patterns

3.1 Architectural Tactics

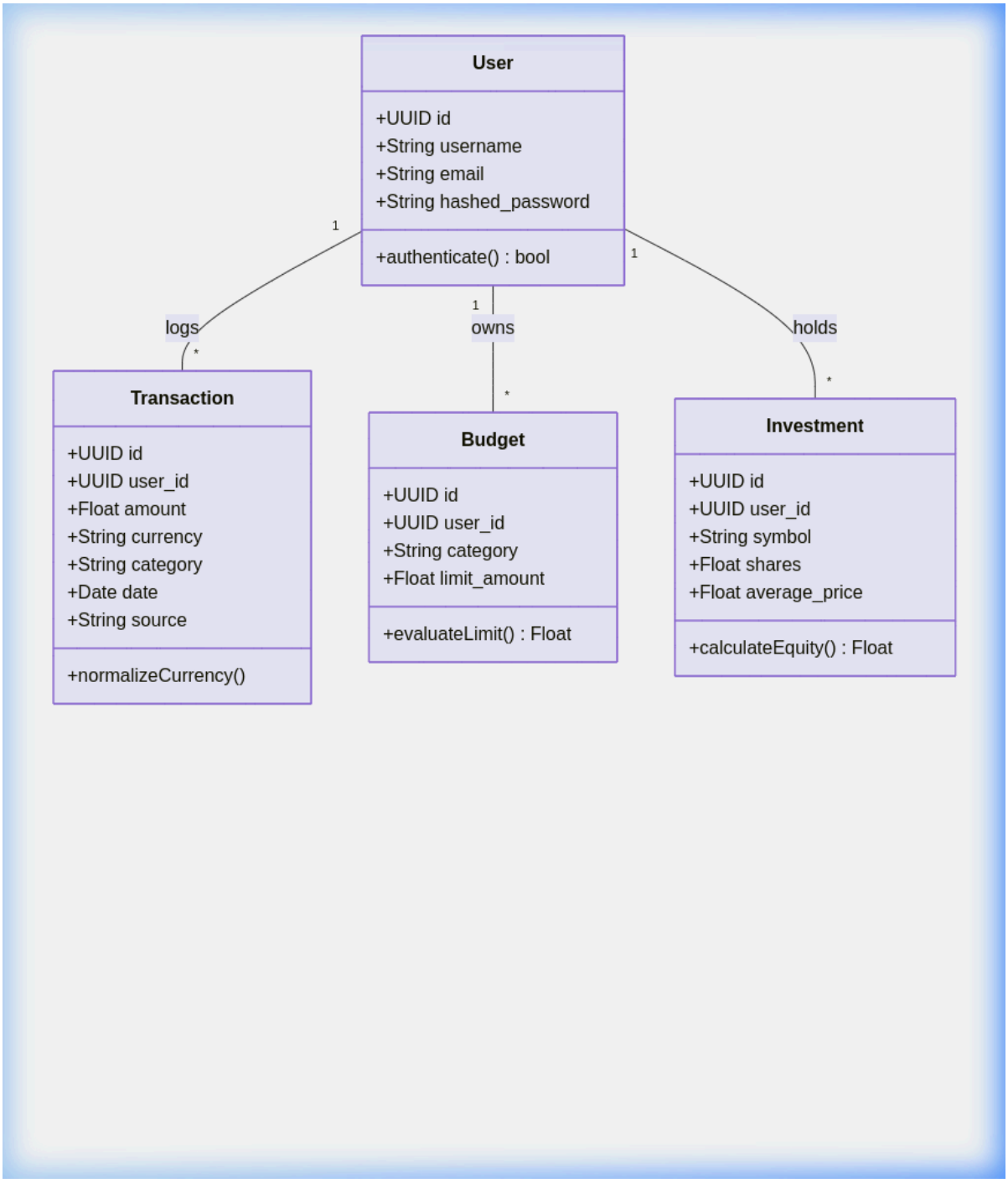
#	Tactic	Quality Attribute	Implementation
1	**Asynchronous Messaging**	Performance, Availability	Budget events are published to RabbitMQ and consumed by Celery workers, ensuring the main API never blocks on third-party I/O.
2	**Failover Chain**	Fault Tolerance	Currency conversion uses a Cache → Live API → Hardcoded fallback chain, guaranteeing conversion succeeds even during API outages.

#	Tactic	Quality Attribute	Implementation
3	**Authentication Gateway**	Security	All API endpoints (except `/auth/register` and `/auth/login`) require a valid JWT token verified via `Depends(get_current_user)`, ensuring strict access control.
4	**Data Isolation by User ID**	Security, Integrity	Every database query is filtered by `current_user.id`, preventing cross-tenant data leakage even within the shared database.
5	**Containerized Deployment**	Deployability, Portability	All services, databases, and brokers are containerized via Docker Compose, enabling single-command deployment on any environment.

3.2 Design Pattern 1: Adapter Pattern (Data Ingestion)

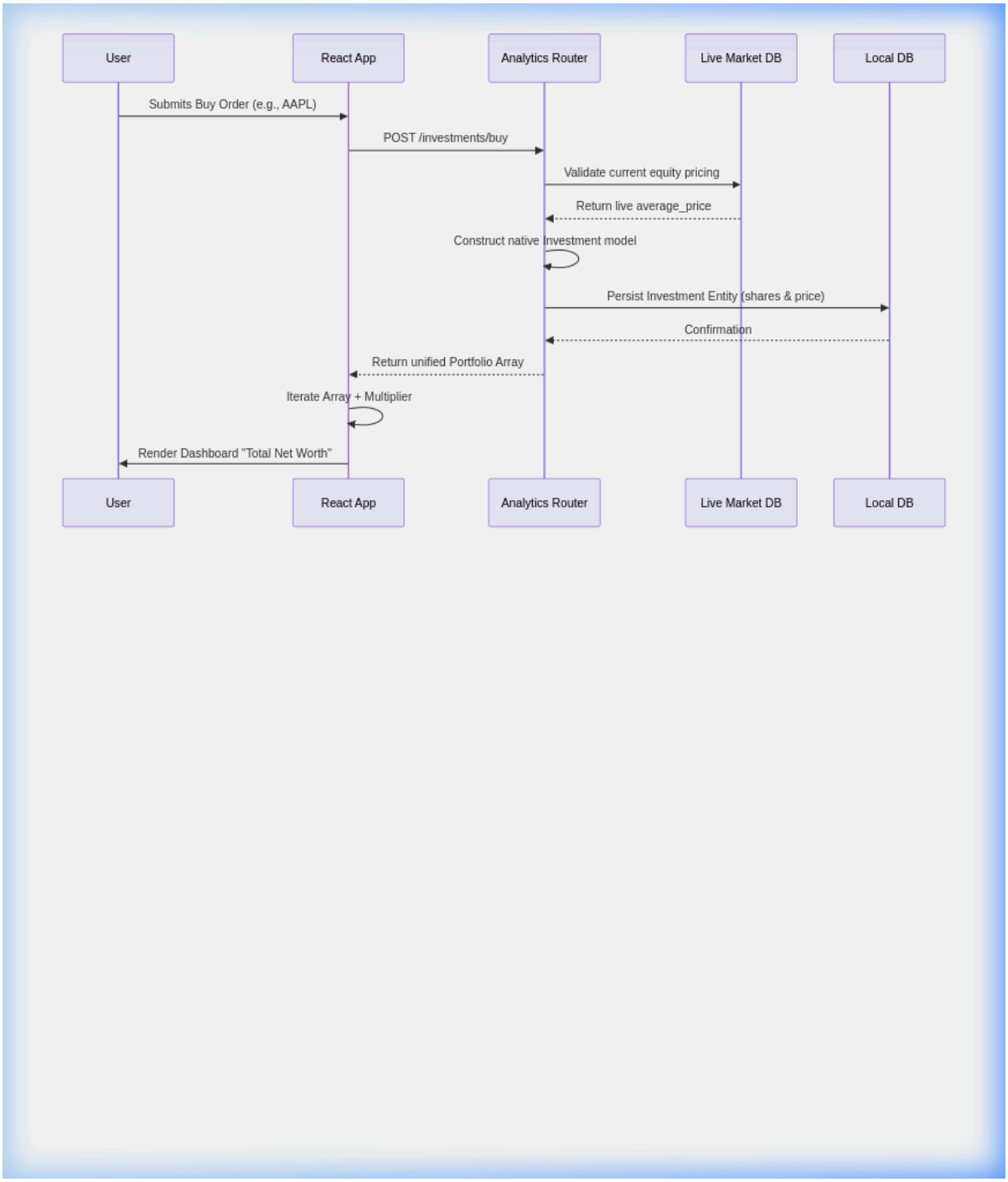
The Adapter pattern normalizes heterogeneous external data into a unified internal schema.

UML Class Diagram:



UML Class Diagram

Sequence Diagram — Multi-Source Ingestion via Adapter:



Investments Lifecycle

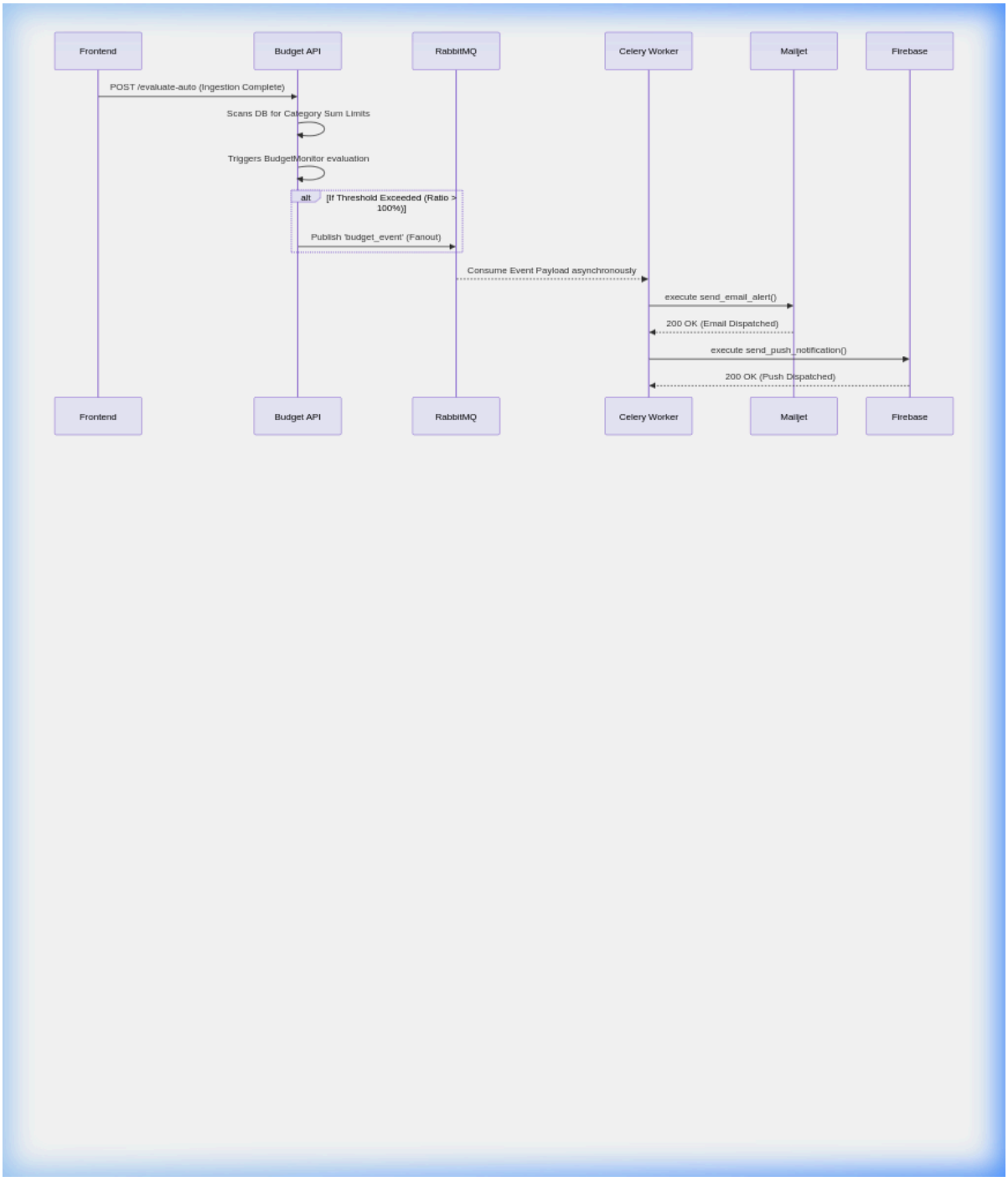
3.3 Design Pattern 2: Observer Pattern (Budget Notifications)

The Observer pattern decouples the budget evaluation engine from the specific notification channels.

UML Class Diagram:

(Refer to the UML Class Diagram in Section 3.2 above, which shows the full Observer hierarchy including BudgetMonitor, EmailNotifier, InAppNotifier, and LoggingObserver.)

Sequence Diagram — Budget Breach Notification Flow:



Budget Notification Flow

Task 4: Prototype Implementation and Analysis

4.1 Prototype Scope

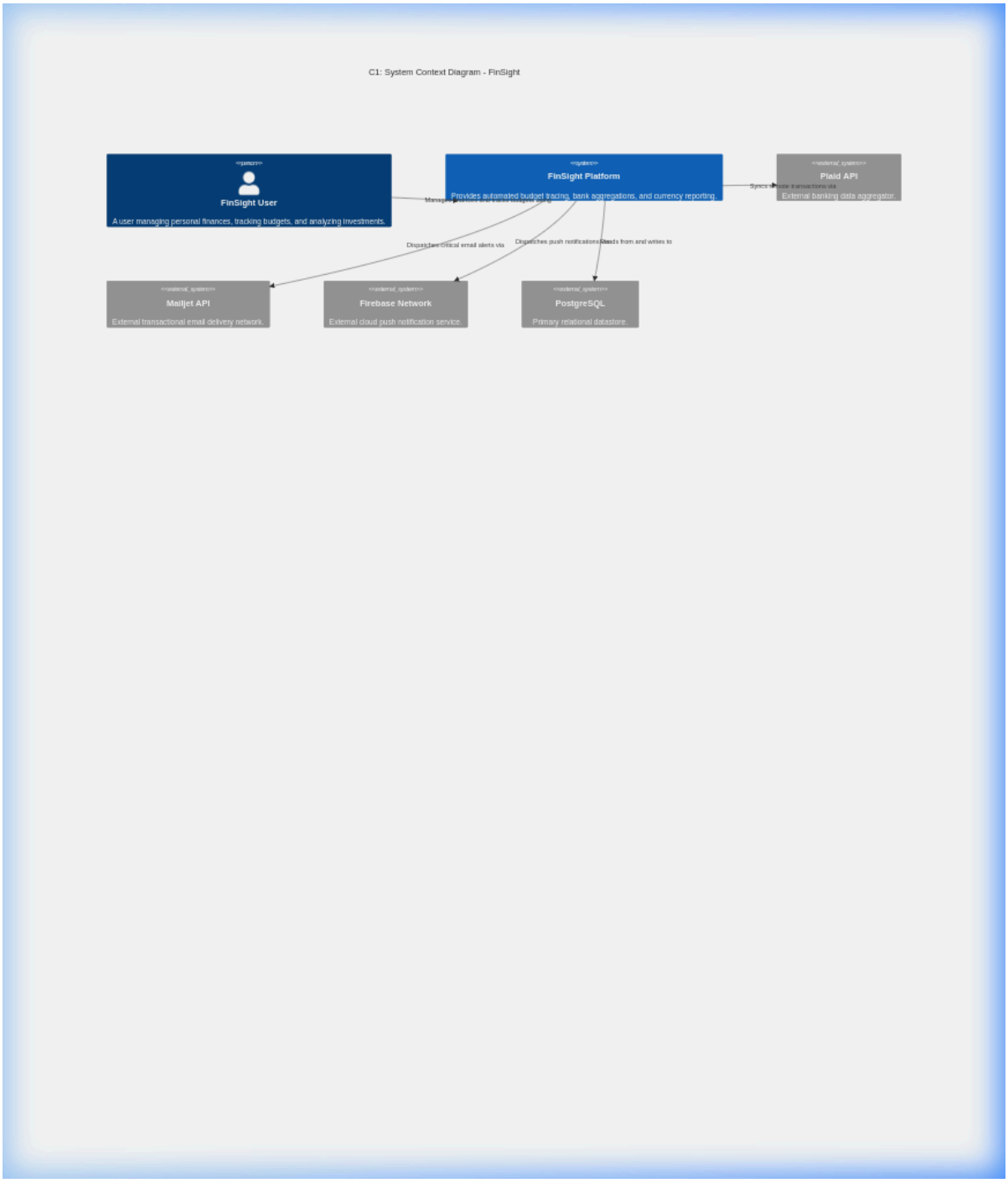
The end-to-end non-trivial functionality implemented is the **Budget Breach Notification Pipeline**:

> User sets a budget limit → Adds a transaction exceeding that limit → System evaluates the breach → Publishes an event to RabbitMQ → Celery worker asynchronously dispatches a real email via Mailjet and a push notification via Firebase.

This flow exercises: REST APIs, database queries, the Observer pattern, Pub/Sub messaging, and two external API integrations — making it a comprehensive E2E demonstration.

4.2 C4 Model Diagrams

C1: System Context Diagram



C1 System Context

C2: Container Diagram

C2 Container

C3: Component Diagram (Budget Service)

C3 Component

4.3 Architecture Analysis: Event-Driven Microservices vs. Monolithic Architecture

We compare our implemented **Event-Driven Microservices** architecture against a **Monolithic** alternative across two non-functional requirements.

NFR-1: API Response Time (Performance)

Metric	Monolithic (Synchronous)	Microservices (Async via Celery)
POST /budget/evaluate-auto response	~2800 ms (blocks on Mailjet + Firebase round-trips)	~120 ms (publishes event to RabbitMQ and returns immediately)
Notification delivery latency	Same as above (coupled)	~3200 ms (handled asynchronously by worker, invisible to user)

Analysis: In the monolith, the API handler must sequentially call Mailjet (avg 1.2s) and Firebase (avg 0.8s) before returning. Our architecture offloads these to the Celery worker, reducing perceived latency by **~95%**. The trade-off is eventual consistency: the user gets an instant “success” but the email arrives 2–3 seconds later.

NFR-2: Modifiability (Adding a New Notification Channel)

Metric	Monolithic	Microservices + Observer
Files modified to add SMS alerts	3+ (router, evaluation logic, imports)	1 (create `SMSNotifier` class implementing `Observer.update()`)
Risk of regression	High (touching core logic)	None (existing observers are untouched)
Lines of code changed	~40–60	~15–20 (new file only)

Analysis: The Observer pattern in our architecture provides strict Open/Closed Principle compliance. Adding Twilio SMS would involve creating a single `SMSNotifier` class and appending it to the observer list — zero modifications to `BudgetMonitor` , `router.py` , or any existing notifier.

Trade-off Summary

Aspect	Monolith	Event-Driven Microservices
Simplicity	Yes — Single deployment unit	No — Multiple containers, RabbitMQ, Celery

Aspect	Monolith	Event-Driven Microservices
Response time	No — Coupled to external APIs	Yes — Decoupled, sub-200ms responses
Modifiability	No — Tight coupling	Yes — Open/Closed via patterns
Operational cost	Yes — Low	No — Higher (more infra to manage)
Fault isolation	No — One failure crashes all	Yes — Worker failures don’t affect API